

Name: _____

Date: _____

PLACE VALUE AND ORDERING INTEGERS, (ROUNDING TO SIGNIFICANT FIGURES)

LO: I can round numbers to a given number of significant figures.

Significant figures

Significant figures (sf) are the meaningful digits in a number. The first sf is the first **non-zero digit**. Count from there.

Leading zeros are not significant. Trailing zeros after a decimal point ARE significant.

Quick examples

●	3456 to 2 sf = 3500	round at 2nd digit
●	0.00347 to 2 sf = 0.0035	leading zeros not significant
●	45.83 to 1 sf = 50	round up
●	0.0034 has 4 sf	only 2 sf (3 and 4)
●	3456 to 1 sf = 3	should be 3000

Key idea

Find the first significant (non-zero) digit. Count the required number of digits from there. Round using the next digit.

3 456 to 1 sf = **3000**

Keep 1 digit, replace rest with zeros

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - large numbers

Round 4728 to 2 sf.

1st sf = 4, 2nd sf = 7

Look at next digit: 2 (less than 5)

Round down: 4700

4700

The 3rd digit (2) tells us to round down.

Example 2 - decimals

Round 0.003856 to 2 sf.

Leading zeros not significant

1st sf = 3, 2nd sf = 8

Next digit = 5, round up: 0.0039

0.0039

Start counting from the first non-zero digit.

Example 3 - one sf

Round 67.3 to 1 sf.

1st sf = 6

Next digit = 7 (5 or more)

Round up: 70

70

One significant figure means one meaningful digit.

Example 4 - trailing zeros

Round 0.05049 to 3 sf.

1st sf = 5, 2nd = 0, 3rd = 4

Next digit = 9 (round up)

0.0505

0.0505

The zero between 5 and 4 IS significant.

YOUR TURN – PRACTICE (deliberate practice)

1. To 1 significant figure

Round to 1 sf.

- a) 47
- b) 832
- c) 5.6
- d) 0.073

2. To 2 significant figures

Round to 2 sf.

- a) 3456
- b) 0.0678
- c) 12.45
- d) 0.005032

3. To 3 significant figures

Round to 3 sf.

- a) 45678
- b) 0.04563
- c) 7.0049
- d) 123456

4. Counting significant figures

How many sf does each number have?

- a) 3.40
- b) 0.005
- c) 2000
- d) 6.020

5. Mixed rounding

Round to the stated accuracy.

a) 56789 to 3 sf

b) 0.003067 to 2 sf

c) 99.6 to 2 sf

d) 0.1005 to 3 sf

e) 5555 to 1 sf

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Leading zeros are not significant ☐

Correct answer: _____

Reason: _____

b) 0.0045 has 4 significant figures ☐

Correct answer: _____

Reason: _____

c) 3.50 has 3 significant figures ☐

Correct answer: _____

Reason: _____

d) 4728 to 1 sf is 4 ☐

Correct answer: _____

Reason: _____

e) The first sf is the first non-zero digit ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A calculator shows **0.00784532**. Round this to: (i) 1 sf, (ii) 2 sf, (iii) 3 sf. Which rounding gives the best balance of accuracy and simplicity?

Expression: _____

Simplified: _____

b) The speed of light is **299,792,458 m/s**. Express this to 1, 2, 3, and 4 significant figures. Scientists often use 3×10^8 . What does this represent in terms of sf?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

